

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		$63 - 1.42 = 61.58$ or $63\,000 - 1\,420 = 61\,580$ (1) Substitution: $\frac{61.58 \text{ ecf}}{63} \times 100$ or $\frac{61\,580 \text{ ecf}}{63\,000} \times 100$ (1) $\% \text{ efficiency} = 97.7[5]$ (1) Alternative (example of a wasted energy route): $\frac{1.42}{63} = 0.0225$ (1) $0.0225 \text{ ecf} \times 100 = 2.25\%$ (1) so $\% \text{ efficiency} = 100 - 2.25 = 97.7[5]$ (1) Alternative (max demand off the winter curve): $55\,000 - 1\,420 = 53\,580$ (1) $\frac{53\,580 \text{ ecf}}{55\,000} \times 100$ (1) So $\% \text{ efficiency} = 97.4$ (1) Alternative (max demand off the summer curve): $(43\,000 \pm 1\,000) - 1\,420 = 41\,580$ (1) $\frac{41\,580 \text{ ecf}}{43\,000} \times 100$ (1) So $\% \text{ efficiency} = 96.7$ (1) Alternative (calculating input energy) Useful energy = $63\,000$ [MW] Input energy = $63\,000 + 1\,420 = 64\,420$ [MW] (1) $\frac{63\,000}{64\,420 \text{ ecf}} \times 100$ (1) So $\% \text{ efficiency} = 97.8$ (1) N.B.1 If in any of the above methods $1\,420$ has not been subtracted then award no marks	1	1 1		3	3	